Raddoppiamento Sintattico in Logical Phonology

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Abstract

I offer an account of Italian *raddoppiamento sintattico* (RS) within the Substance-Free Logical Phonology (LP) framework. By relying on set-theoretic formalisms, RS is analyzed as an instance of compound unification, wherein a lexically-specified but phonologically-underspecified (empty) segment ($X[\emptyset]$) is filled by the features of a following consonant. It moves RS away from moraic or morphological accounts and back to the domain of phonology, including in cases of blocking (e.g., phonetic pausing, glottal stop insertion, and vowel lengthening). I show that both the application and non-application of RS emerge merely from evaluating whether the empty segment is available for unification and ostensible blocking simply falls out from a lack of a subtraction rule preceding unification.

Keywords: *Raddoppiamento sintattico*, Logical Phonology, underspecification

1 A feature-filling approach

Italian *raddoppiamento sintattico* (RS) is a morphophonological process that causes the initial consonant of a word to geminate (or "double") when preceded by certain triggers, such as certain monosylables (e.g., the preposition *a* in *a casa* $\rightarrow$ [akkaza] 'at home') or certain oxytones (e.g., *farà bene* $\rightarrow$ [faˈrabbɛːne] 'S/he'll do well'). Accounts of RS in Prosodic Phonology (Nespor & Vogel, 1986) model the phenomenon as a domain-bound process within the phonological phrase. However, in spontaneous speech, RS is frequently blocked by factors such as phonetic pausing, sudden pitch movement, vowel lengthening, and glottal stop insertion (Absalom & Hajek, 2006). While prosodic accounts stipulate morphosyntactic markers or domains to accommodate these patterns, Substance-Free Logical Phonology (LP) provides a segmental alternative (noteably not even appealing to moraic constraints also disabused of in Absalom et al., 2002.) The present work does not necessarily offer a new story of the phenomenon, but rather restricts the existing story to phonology via an LP account collapsing triggering and blocking under the same set-theoretic logic defined for mutability by underspecficiation (cf. Inkelas and Cho, 1993).

2 Background on Logical Phonology

As the present work is a sort of squib, the reader unfamiliar with LP is encouraged to consult Bale et al. (2014, 2019) and Gorman and Reiss (2025) for the core tenets and applications of the theory. Most relevant to the story of RS is the fact that LP models phonological objects using set theory. Segments are sets of valued features; natural classes are sets of such sets. Germaine to RS is how gemination can be thought of as filling in an empty space with a fully specified segment. Here, feature-filling processes involve unification ($\sqcup$). Bale et al. (2019) define unification as follows:

$$A \sqcup B = A \cup \{\alpha F \mid \alpha F \in B \wedge -\alpha F \notin A\} \tag{1}$$

The definition in (1) states that unifying two feature specifications yields their union, constrained by consistency, meaning that unification fails if it would result in a set containing conflicting features $+F$ and $-F$. If A possesses a contradictory feature value to an attribute in B (e.g., A contains $[-\text{Voice}]$ while B contains $[+\text{Voice}]$), unification is undefined. When unification fails due to inconsistencies, the mapping defaults to the identity function, leaving the segment unaltered.

The sole segment that permits unconstrained unification is the maximally unspecified segment, which lacks any featural content. While Bale et al. (2019) identify both curly brackets ({}) and the slashed circle (∅) as valid notations for the empty set, I will use ∅ throughout. Crucially, LP distinguishes between a featureless segment attached to a timing slot, denoted here as $X[\emptyset]$ (or simply ∅ for readability), and true nothingness or the total absence of a segment, which is represented by ϵ following Bale and Reiss (2018). Because ∅ is maximally underspecified, it avoids featural mismatch and is uniquely capable of undergoing full unification, producing what looks like the doubling of an adjacent consonant.

This set-theoretic perspective provides an alternative to previous structural analyses of raddoppiamento sintattico. The present work follows Borrelli (2002) in analyzing RS by projecting empty skeletal X-slots at the right edge of RS-inducing words, but diverges in motive (a note on the historical motivation will be provided in the discussion). Like Borrelli, I hold a segment to be an X-slot associated with a set of features (Reiss, 2026). However, rather than relying on prosodic weight constraints to dynamically project an empty slot, the story here simply assumes the empty set (or empty segment) is present in the UR without requiring special morphological or prosodic stipulations. This approach treats the application or lack of application of RS under a single evaluation of whether the empty segment ∅ is present and available for unification.

3 Lexical representations and inalterability

Italian distinguishes "catalytic" triggers, which induce RS (e.g., *tre* 'three'), from "quiescent" words, which never cause RS (e.g., *due* 'two'). Following the principle of Inalterability as Prespecification (Gorman & Reiss, 2025; Inkelas & Cho, 1993), catalytic triggers end with an unassociated X-slot, stored lexically as ∅; thus, to account for RS, it appears the UR of 'three' is [tre∅]. Quiescent words lack this empty segment, meaning they do not have the right environment for the unification rule developed below to apply. They are thus, inalterable due to their prespecification of features, whereas the empty segment is wholly underspecified. Confusingly, the words seem to be triggers since they present the environment for a rule to apply, but it's also the words that yield the target of the feature-filling, which is the empty segment.

Also relevant are non-monosyllabic triggers, such as paroxytones—words stressed on the penultimate syllable, such as *come* 'how', *dove* 'where', and *qualche* 'some'. The presence of ∅ in the UR of these specific paroxytones accounts for their catalytic behavior without relying on any active stress-based rule.

Some words with final stress (e.g., *città* 'city') also induce RS. I propose that such words inherently possess an underlying ∅ at their right edge, so the UR for 'city' is /tʃitta∅/. This eliminates the need for a synchronic rule that tags such words or adds an extra gemination trigger, assuming that speakers actually have the empty segment "baked in" to all triggers in the lexicon. This is practically different from an insertion rule, which would be formulated by targetting true nothingness (denoted by ϵ, Gorman and Reiss, 2025) instead of the segment that is totally underspecified but still linked to an X-slot in the UR (∅). In Gorman and Reiss' insertion account, there's an X-slot added, but for RS, which is, in the case of unstresesd monosyllables and some oxtyones, lexically-specified (Absalom & Hajek, 2006), it is more sensible to assume the UR of triggers already contains the empty segment.

The underlying ∅ can be filled by geminating the following consonant (RS), or, if RS fails to apply, by geminating the previous vowel, although it's an empirical/probabilistic puzzle as to when each occurs.

4 Compound unification

RS is the doubling of a consonant across word boundaries, and because it operates across these boundaries, the rule must be a late, postlexical process, which is why the phenomenon is termed 'syntactic' doubling (*raddoppiamento sintattico*) rather than mere phonological gemination. Since it appears as surface copying of a whole segment (or feature bundle), I model RS as total assimilation. To conform to the Singleton Set Restriction (Bale et al., 2014, 2019), total assimilation is defined as compound unification, an iterative sequence of rules unifying one feature at a time. This sequence can be summarized using the standard LP variable shorthand for multiple consecutive unification rules (Reiss, 2024):

$$[\,] \sqcup \begin{Bmatrix} \alpha_1 F_1 \\ \alpha_2 F_2 \\ \vdots \\ \alpha_k F_k \end{Bmatrix} \;/\; \underline{\quad} \begin{bmatrix} +\textsc{Cons} \\ \alpha_1 F_1 \\ \alpha_2 F_2 \\ \vdots \\ \alpha_k F_k \end{bmatrix} \tag{2}$$

The shorthand in (2) targets the universal natural class, $[\,]$. If the target segment is fully or partially specified, the rule is applied vacuously. If the target contains a conflicting feature, unification is undefined and defaults to an identity mapping. Following Bale et al. (2019), when unification is undefined, the application is vacuous, leaving the target unaltered. The only non-vacuous and genuine feature-filling case occurs when the target is $\emptyset$. While I do not delve introduce the operator here, the real reason that it is not the case that every $[+\text{CONS}]$ gets switched to $[-\text{CONT}]$ regardless of it's original specification is that LP has no simultaneous swap operator; the prespecified $[+\text{CONT}]$ of, say, an /s/ would have to be set-subtracted with a different operator and rule. This is because unification again only applies when the output would be consistent, i.e., have no conflicting features.

Another potential hurdle to overcome in the present analysis is how to prevent unassociated $\emptyset$ segments from surfacing. An easy solution may be to introduce the aforementioned set-subtraction operator, but as Reiss (2025) shows, it is a logical impossibility to write a segment deletion rule that targets an empty or underspecified segment without also deleting everything more specified than it. Because $[\,]$ is the universal natural class, it denotes any segment that is a superset of the empty set, i.e., every segment in the language. A rule targeting it would therefore delete every single consonant and vowel that occurs before a vowel or a pause.

Fortunately, a deletion rule is unnecessary. As LP is interpreted within Cognitive Phonetics (Reiss & Volenec, 2022), features are transduced directly into articulatory motor programs. If a segment arrives at the interface as a completely empty set of features, there are simply no articulatory instructions to execute. It naturally surfaces as a null phonetic event (silence).

5 Search, change, and blocking

Raddoppiamento sintattico can be characterized via a Search and Change sequence (Dabbous et al., 2024). To ensure strict adjacency, the Terminator must be minimally specified, and the featural requirements placed in the Condition parameter:

Initiator (INR): $[\,]$. The parse initiates across the universal natural class.

Direction (DIR): Right. Unification seeks features rightward.

Terminator (TRM): $[\,]$. The probe halts at the adjacent segment.

Condition (COND): $[+\text{CONS}]$. The change is licensed iff the TRM is a consonant.

Change (OUT): The output fills the INR variable ($\emptyset$) with the features of the TRM via compound unification.

Formulating the search in this way provides a mechanism for ostensible blocking effects. Here, I consider three RS blockers documented by Absalom et al. (2002).[1] The first blocking scenario is that of phonetic pausing. The authors note that "when

[1] I leave the question of sudden pitch movement as a blocker unaddressed.

a pause (however slight) occurs at the word boundary between word$_1$ and word$_2$ in RS contexts, doubling due to RS does not happen". Rather than positing a set-subtraction rule for prepausal deletion of ∅, this blocking follows from the Search and Change parameters. If an underlying ∅ is followed by a pause (e.g., [fa'ra // bɛːne]), the strict rightward search (TRM: [ ]) terminates on the pause itself. Because a pause fails to satisfy the condition of being [+Cons], the change is unlicensed and unification is entirely blocked.

A second blocker, glottal stop insertion, operates on a different LP mechanism: unification failure. Absalom et al. treat pausing and glottalization as distinct phonetic events, noting that glottal stops can fill the boundary slot independently. The authors entertain the possibility that the glottal stop is a mora-filling device, but regardless of its origin, once the glottal stop fills the word-final position, the slot is rendered inalterable to the adjacent consonant features. As it fills the word-final position in an example like [an'doʔ bɛːne] 'it went well', the ∅ segment is filled with glottal features. This creates a featural clash that causes the subsequent right-to-left RS compound unification to fail, thereby blocking the adjacent consonant from doubling.

The final blocking scenario is that of final vowel lengthening. It is well-known that geminate consonants (including RS) and long vowels do not cooccur (Borrelli, 2002; Colombo et al., 2024). The present work acknowledges that weight may condition this restriction, but assumes that RS and vowel lengthening compete for the underlying ∅ slot of oxytones. This establishes a doubly-bleeding relationship: if left-to-right vowel spreading successfully unifies into the empty slot first, the slot becomes featurally specified. Subsequent right-to-left RS unification fails due to featural mismatch. Thus, *andò bene* 'it went well' can either be [an'doːbɛːne] or [an'dobbɛːne], but never **[an'doːbbɛːne].

The existence of both forms in variation is treated passingly by Absalom et al. (2002) as an EVAL tie in Optimality Theoretic terms, yielding two equally optimal candidates. However, within a strictly rule-based formalism, positing simultaneous or competing rules is absurd. Perhaps we can account for this variation using the Blueprint Model of Production (Nelson, 2024). Nelson demonstrates that optionality in sandhi processes emerges not from optional rules, but from variations in speech planning chunks. If *andò* and *bene* are processed within the same planning chunk, the RS Search and Change algorithm targets the adjacent consonant, applying compound unification and bleeding vowel lengthening. Conversely, if the speaker parses the words into separate planning chunks, the rightward RS search cannot cross the chunk boundary. The condition for RS is unmet, and default left-to-right vowel lengthening fills the X-slot instead. This approach derives both surface variants through the composition of phonological rules and different planning parses.

6 Sample derivations

Now, consider the following derivations presented first with a sequential unification ruleset and then abbreviated with segment mapping diagrams (SMDs, Reiss, 2024). TRM segments are abbreviated as $[+\text{Cons}]$.

1. **Lexical trigger before consonant (*a casa* $\rightarrow$ [akkaza]):**
   The input is /a∅kaza/. Because the Singleton Set Restriction forbids copying multiple features in one step, total assimilation is formalized as compound unification ($\bigsqcup\{\alpha_i F_i\}$):

UR	/a∅kaza/	
F_1: $[\,] \sqcup \{+\textsc{Cons}\}$	a$\{+\textsc{Cons}\}$kaza	(k is $+\textsc{Cons}$)
F_2: $[\,] \sqcup \{-\textsc{Cont}\}$	a$\left\{\begin{matrix}+\textsc{Cons}\\-\textsc{Cont}\end{matrix}\right\}$kaza	(k is $-\textsc{Cont}$)
F_3: $[\,] \sqcup \{+\textsc{Dors}\}$	a$\left\{\begin{matrix}+\textsc{Cons}\\-\textsc{Cont}\\+\textsc{Dors}\end{matrix}\right\}$kaza	(k is $+\textsc{Dors}$)
F_4: $[\,] \sqcup \{-\textsc{Voice}\}$	a$\left\{\begin{matrix}+\textsc{Cons}\\-\textsc{Cont}\\+\textsc{Dors}\\-\textsc{Voice}\end{matrix}\right\}$kaza	(k is $-\textsc{Voice}$)
$\vdots$	$\vdots$	
SR	[akkaza]	

2. **Lexical trigger before vowel (*a Empoli* $\rightarrow$ [a empoli]):**
   The input /a∅empoli/ encounters the $[-\textsc{Cons}]$ specification in /e/, so unification (2) fails constraint checking on $[+\textsc{Cons}]$. The segment successfully passes through the derivation unaltered since an empty feature bundle provides no articulatory motor programs to execute.

$$\begin{array}{lccc} \text{UR} & a & X[\emptyset] & e \\ & \downarrow & \downarrow & \downarrow \\ \text{SR} & a & X[\emptyset] & e \end{array}$$

3. **Quiescent word before consonant (*due cani* $\rightarrow$ [due kani]):**
   The quiescent /due/ lacks a trailing ∅; the final segment is inalterable due to prespecification.

$$\begin{array}{lcccccccc} \text{UR} & d & u & e & & k & a & n & i \\ & \downarrow & \downarrow & \downarrow & \text{(no target present)} & \downarrow & \downarrow & \downarrow & \downarrow \\ \text{SR} & d & u & e & & k & a & n & i \end{array}$$

4. **Blocking via glottal stop (*andò bene* $\rightarrow$ [an'doʔbɛːne]):**
   The input /ando∅bɛne/ undergoes glottalization (perhaps due to moraic requirements). Because glottal features occupy the empty slot, RS compound unification ($\bigsqcup\{\alpha_i F_i\}$) cannot apply due to inconsistency.

$$\begin{tikzcd}
\text{UR} & o \arrow[d] & X[\emptyset] \arrow[d, "\sqcup\{+\text{C.G.}\}"'] & b \arrow[d] \arrow[dl, dashed, "\sqcup\{\alpha F\}\ \text{(fails)}"] \\
\text{SR} & o & \text{ʔ} & b
\end{tikzcd}$$

5. **Blocking via phonetic pause (*farà // bene* $\rightarrow$ [fa'ɾa // bɛːne]):**
   The underlying empty segment is followed by a pause ($//$): /fara∅ // bɛne/. The strict rightward search terminates on the pause (cannot look past current planning chunk). Because the pause lacks $[+\textsc{Cons}]$ features, it fails the COND parameter, blocking the unification rule. As discussed in section (4), an empty segment gets no motor interpretation when SR passes through phonetics.

$$\begin{tikzcd}
\text{UR} & a \arrow[d] & X[\emptyset] \arrow[d, "\text{(search fails)}"] & // \arrow[d] & b \arrow[d] \\
\text{SR} & a & \emptyset & // & b
\end{tikzcd}$$

6. **RS Bleeds Vowel Lengthening (*città bella* $\rightarrow$ [tʃittabbella]):**
   The input /tʃittà∅bella/ has an underlying ∅ at the end of the oxytone. If RS compound unification ($\bigsqcup\{\alpha_i F_i\}$) applies before vowel lengthening, it right-to-left unifies the features of /b/ into the ∅. This fills the slot by copying all features from the adjacent consonant iteratively, cleanly bleeding any subsequent vowel lengthening.

$$\begin{tikzcd}
\text{UR} & a \arrow[d] \arrow[dr, dashed, "\text{(bled)}"'] & X[\emptyset] \arrow[d] & b \arrow[d, "\sqcup\{\alpha_i F_i\}"'] \arrow[dl] \\
\text{SR} & a & b & b
\end{tikzcd}$$

7 Discussion and conclusion

While this paper should be considered a synchronic phonological model, the formalization of the *raddoppiamento sintattico* target as ∅ (really $X[\emptyset]$) reflects the phenomenon's historical origins. As established by Loporcaro (1988, 1997) and Borrelli

(2002), RS originated in Late Latin as a process of total regressive consonant assimilation (e.g., AD CASAM > *accasa*). With the final consonants' identity so seldom surfacing (since a preposition like *AD* necessarily precedes a noun), they were subsequently lost through phonological erosion, leaving behind their structural timing slots devoid of featural content, or in set-theoretic terms, the empty set was all that remained in the URs of such words. Therefore, the synchronic compound unification rule proposed here is simply the modern reflex or rephonologization of total assimilation, in this case, feature-filling of the empty segment. As a late, postlexical rule that still applies across word boundaries, the phenomenon retains its designation as *syntactic* doubling. It's not improbable however that lexicalized compounds (like *chissà* 'who knows' or *davvero* 'truly'), are no longer derived synchronically via $X[\emptyset]$ unification but rather stored with underlying geminates.

Furthermore, while this model derives the blocking effects of phonetic pausing, glottal insertion, and vowel lengthening through strict adjacency constraints and unification failure, I sidestep the sudden pitch movement blocker in Absalom and Hajek (2006) and Absalom et al. (2002). Future research may determine that pitch is actually a clue as to chunking, and the real blocker being chunk separation, not pitch shift.

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